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Towards valorization of rice straw into bioethanol and lignin: Emphasizing critical role of deep eutectic solvent components in biorefining process

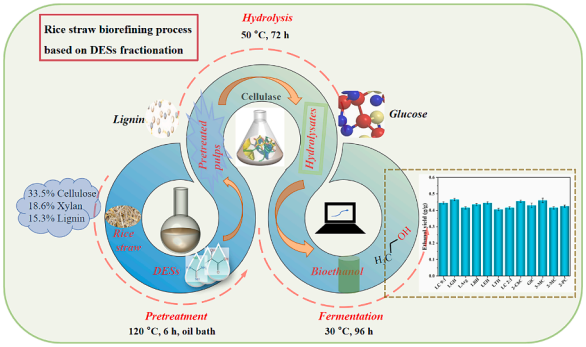
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HIGHLIGHTS

- Structures of DESs affected the pretreatment efficiency and ethanol fermentation.
- Lactic acid/guanidine hydrochloride effectively removed lignin and xylan.
- The hydrolysates of various DES pulps had impressive ethanol yields.
- 302.75 g ethanol and 118.98 g lignin were harvested from 1 kg dried biomass.

GRAPHICAL ABSTRACT



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ABSTRACT

Deep eutectic solvents (DESs) offer a potential opportunity in biomass utilization industries. This work emphasized the impact of hydrogen bond donors (HBD) and acceptors (HBA) on deconstruction and valorization of rice straw. Acidity, alkyl chain length, hydrogen bonding ability and functional groups of HBD and HBA appeared to be important factors affecting the fractionated pulps and lignins, which further influenced ethanol fermentation. Among the candidate DESs, lactic acid/guanidine hydrochloride (LGH) was proved to be the most suitable one due to the excellent delignification and xylan removal. For the downstream fermentation process, 0.47 g g^{-1} of bioethanol with 0.55 g/L h^{-1} of productivity can be obtained from the LGH pulp's hydrolysate. Mass balance showed 302.8 g bioethanol and 119.0 g technical lignin can be co-generated from 1 kg dried rice straw. This "green" valorization strategy offers a promising scheme in biorefinery of lignocelluloses.

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1. Introduction

The unsustainable supplementation of the fossil resources and environment pollution forced continuous exploration of the alternative renewable resources for sustainable chemicals and materials (Singh et al., 2022). Over the past decades, numerous researches have been devoted to construct a suitable biomass fractionation strategy for obtaining high -value added products from lignocellulosic biomass, which is the most abundant renewable bioresource on earth (Gao et al., 2023). In recent years, biomass fractionation based on deep eutectic solvents (DESs) has received increasing attention due to properties such as low vapor pressure, low cost and biodegradability, recyclability, tunability and simplicity for preparation (Cao & Su, 2021). The typical DESs consist of two or more hydrogen bond donor (HBD) and hydrogen bond acceptor (HBA) through a certain molar ratio (del Mar Contreras-Gámez et al., 2023). It has been widely reported that the lignin-carbohydrate complex of biomass can be effectively dissociated in DESs accompanying with delignification, resulting in cellulose enriched pulps and a lignin stream with largely preserved active ether bonds and functional groups (Cheng et al., 2023). In the current research, efforts have been made to explore the relationship between the structure and performance of DESs in biomass pretreatment. Acidity, hydrophilic ability, polarity, alkyl chain length, functional groups and molar ratio of HBA/HBD have been considered as important factors affecting the solvents performance (Xu et al., 2021b). HBD was mostly reported to play the main role in biomass solubility as compared to HBA (Ji & Lv, 2020), while the chloride anion of choline chloride (HBA) was also proved to be active component for cleavage of β-O-4 (Smink et al., 2019). Particularly, organic acids such as lactic acid, glycolic acid are commonly used HBD for effective biomass pretreatment, and choline chloride, the typical ammonium salts, are usually adopted as HBA to built DESs (del Mar Contreras-Gámez et al., 2023; Duan et al., 2023). Additionally, the free protons in DES and molecular chain length of HBD were proposed to affect the delignification efficacy (Xu et al., 2021c). However, to get better understanding of the structure–function relationship, the type of DESs should be expanded to more ammonium salts (HBAs) and organic acids (HBDs) with different functional groups. Moreover, the effects of the HBA/HBD structures of DESs on the further valorization of the pulps have not been addressed, which is the decisive issue about the feasibility of DESs-based biorefineries.

The molar ratio of HBA to HBD is critical for building DES with good properties for biomass deconstruction (Francisco et al., 2013). Considering the fact that all the selected candidates could not form satisfactory liquid in the same molar ratio,when using lactic acid as the counterpart (HBD), 1:9 of molar ratio was applied; while 1:2 was used when choline chloride was set as the constant counterpart (HBA). On this basis, in this work, the effect of HBA/HBD structures on the deconstruction performances was investigated, the correlations of the structural properties of DES towards saccharafication and lignins fractionation were clarified. Furthermore, the critical role of HBA/HBD structures of DESs in the downstream valorization processes (further conversion of the pulp into bioethanol) was also emphasized.

2. Materials and methods

2.1. Materials

Rice straw was collected from local fields in Guangdong Province. After drying out, the straw was smashed into 250–400 μm particles and stored in a vacuum desiccator before use. The chemicals (>98 wt%) used for DESs preparation were listed in Table 1 and purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. (Shanghai, China). Other chemicals under analytical grade (including glucose, tryptone, yeast extract, malt extract, and acetone, triethylenediamine) were supplied by Macklin and Sigma-Aldrich. Cellulase/Xylanase from *Trichoderma reesei* was purchased from Sigma-Aldrich (St. Louis, MO, USA) and used as received. The *Saccharomyces cerevisiae* YSX4C222, genetically engineered and adapted to efficiently utilize glucose and xylose, was preserved in laboratory.

Table 1
Abbreviations of different DESs with HBA/HBD molar ratios of 1:9 and 1:2.

Entry	DESs			
	HBA	HBD	Molar ratio of HBA/HBD	Abbreviation
1	Choline chloride	Lactic acid	1:9	LC 9:1
2	Guanidine hydrochloride	Lactic acid	1:9	LGH
3	Arginine	Lactic acid	1:9	LArg
4	Betaine hydrochloride	Lactic acid	1:9	LBH
5	Ethanolamine hydrochloride	Lactic acid	1:9	LEH
6	Triethylamine hydrochloride	Lactic acid	1:9	LTH
7	Choline chloride	Lactic acid	1:2	LC 2:1
8	Choline chloride	2-Chloropropionic acid	1:2	2-ChC
9	Choline chloride	Glycolic acid	1:2	GLC
10	Choline chloride	3-Mercaptopropionic acid	1:2	3-MC
11	Choline chloride	2-Mercaptopropionic acid	1:2	2-MC
12	Choline chloride	2-Phenylpropionic acid	1:2	2-PC

2.2. DESs preparation and rice straw fractionation

The DESs were prepared by mixing HBA and HBD in a molar ratio of 1:9or 1:2. The detailed information and abbreviations were listed in Table 1. Then, the mixtures were heated and stirred at a temperature of > 80 °C in a closed flask until a homogenous colorless solutions formed.

As for fractionation of rice straw, DESs were mixed-well with the rice straw powder at 10 wt% of solid loading, followed by stirred at 120 °C for 6 h in oil bath. At the end of pretreatment, the mixture was separated by centrifugation. The solid residues were thoroughly washed with anhydrous ethanol and then deionized water before freeze-drying. Ethanol washing solution was combined to centrifugal supernatant, and then concentrated by rotary evaporation to remove ethanol. Then, 3 times volume of deionized water was added and the pH was adjusted to 2–3 to precipitate dissolved lignin. After placed at 4 °C for 48 h, the precipitation (recorded as Lignin-rich materials (LRM)) were collected by centrifugation (10000 rpm, 15 min), and lyophilized after thoroughly washing by deionized water (pH 2–3). The precipitated lignin samples were obtained and recorded accordingly, which were: L-LC 9:1, L-LGH, L-LArg, L-LBH, L-LEH, L-LTH, L-LC2:1, L-2-ChC, L-GLC, L-3-MC, and L-2-MC.

2.3. Saccharification and ethanol fermentation

The saccharification was initially implemented with biomass loading of 2.86 % (w/v) and cellulase dosage of 40–80 FPU/g glucan at 50 °C, and the duration was 72 h. Aliquot samples were withdrawn at specified time intervals for high performance liquid chromatography (HPLC) analysis (Hou et al., 2018). For high titer ethanol fermentation, the enzymatic hydrolysis step was also conducted at a relatively high solid loading (15 %, w/v) with the cellulase dosage of 15 FPU/g glucan in a 50 mL of conical flask (with 30 mL of working volume), with a rotation rate of 250 rpm. All experiments were repeated at least 3 times.

The enzymatic digestibility of cellulose and xylan were calculated as follow:

$$\text{Polysaccharides digestibility} = \frac{\text{Released sugar weight}}{\text{Overall polysaccharides weight in pulp}} \times 100\% \quad (1)$$

Bioethanol fermentation was conducted using the enzymatic hydrolysates of the DESs pulps. The *Saccharomyces cerevisiae* YSX4C222 that could effective co-metabolism of xylose and glucose in lignocelluloses hydrolysate was adopted as the bioethanol producing strain (Wu et al., 2023). The seed medium for YSX4C222 consisted of 20 g/L glucose, 10 g/L yeast extract, and 20 g/L tryptone.

Before inoculation, additional nutrients were added into the hydrolysates (3 g/L yeast extract, 3 g/L of malt extract, and 5 g/L tryptone) according to the conditions in literature (Zhang et al., 2019), and the pH of the hydrolysates were adjusted to 5.5 by ammonium hydroxide before inoculation. After autoclaving at 121 °C for 20 min and cooling down to the room temperature (~20 °C), the active YSX4C222 strains were inoculated with a rate of 10 % (v/v). Batch fermentation was performed in conical flasks. The temperature and rotation rate were 30 °C and 200 rpm, respectively. Samples were collected periodically, and the bioethanol and residual sugars concentration were determined by HPLC and gas chromatography (GC), respectively. All the fermentations were repeated in triplicates.

2.4. Assay

The cellulose, xylan, lignin and ash contents of the samples were determined according to the standard NREL analytical procedure (Sluiter et al., 2008). The sugars in enzymatic hydrolysate and the fermentation broth were monitored using HPLC method according to the previous report (Hou et al., 2018). Bioethanol concentration was determined by GC as described previously (Wu et al., 2023).

As for the characterization of pulps and lignin specimens after DESs fractionation, the crystallinity of pulp was examined by X-ray diffractometer (XRD, Rigaku SmartLab 9 kW, Japan) and fourier transforminfrared spectrometer (FTIR, Thermo-Fisher Nicolet 6700, US) which were conducted as the method used in previous report (Hou et al., 2018).

The structural information and the functional groups of lignin were analyzed and determined by two dimensional ^{1}H–^{13}C heteronuclear single-quantum coherence (2D-HSQC) NMR and ^{31}P NMR techniques (AVANCE III HD 600 MHz, German). Semi-quantitative analysis was performed according to the method in literature (Chen et al., 2020). The molecular weight of the lignin samples was measured using high performance gel permeation chromatography (GPC, Waters 1525, US) (Wu et al., 2023).

3. Results and discussion

3.1. Effect of DESs fractionation on composition of rice straw

Lots of factors, including acidity, alkyl chain length, functional groups, molar ratio of HBA/HBD, hydrophilic ability, polarity and hydrogen bond acidity/basicity (Kamlet Taft parameters) were reported to be critical for influencing their pretreatment performance (Xu et al., 2021a). In order to explore the structure–function relationship, lactic acid based DESs that consist of various HBAs and choline chlorides based DESs containing different HBDs were employed to fractionate rice straw. Chemical compositions of pulps after DESs fractionation are shown in Fig. 1a.

For lactic acid based DESs, the contents of xylan and lignin were decreased significantly as compared to the raw rice straw, except for the L-Arg involved case. This is also responsible for the lowered solid recovery (47.1–76.4 wt%) and increased cellulose contents (49.2–67.9 wt % vs 33.5 wt%). It was found that low contents of xylan and lignin were mostly observed for the pulps containing high cellulose content. Hence, it appears that these DESs could remove large amounts of amorphous substances, such as hemi-cellulose and lignin (Fig. 1b), thus receiving the cellulose-enrich pulps.

The different chemical compositions of the pulps were resulted from the structure changes of HBA. LArg possessed the weakest deconstruction power due to the lowest removal rate of xylan and lignin. The specific structure and high viscosity of LArg may account for the poor performance (Fig. 1b) (Xie et al., 2023). Structurally, arginine has the longest carbon chain that containing the electron-donating groups such as amino group and hydroxyl group, which make it prefer to form complex intra-molecular hydrogen bonding network with lactic acid. Consequently, it is highly viscous and requires extra energy to break down the interactions within the solvent, which led to the weakest interaction with the lignocellulosic matrix and the poorest decomposition efficacy. As expected, LGH exhibited an excellent capability of biomass deconstruction for the decrease of xylan and lignin contents in the pulp, which was possibly due to the presence of more electron-donating groups and the strong acidity of the solvent (Hou et al., 2018). The mechanism of LGH pretreatment have been proposed by theoretical calculation. It was proved that Cl^- played a dominant role in the dissolution of hemicellulose, and the guanidine group in GuHCl with the delocalized π bond strength the coordination ability of Cl^-(Zhu et al., 2023). As for the performance of LC and LBH, with similar structure of HBA, both of them contain the quaternary ammonium ions. However, betaine hydrochloride has a carboxyl group (COOH) while choline chloride contains –OH group. Two hydrogen bonding sites (O = and –OH) of LBH make it easier to form hydrogen bonds than LC. Consequently, higher xylan removal rate can be realized than when using LBH

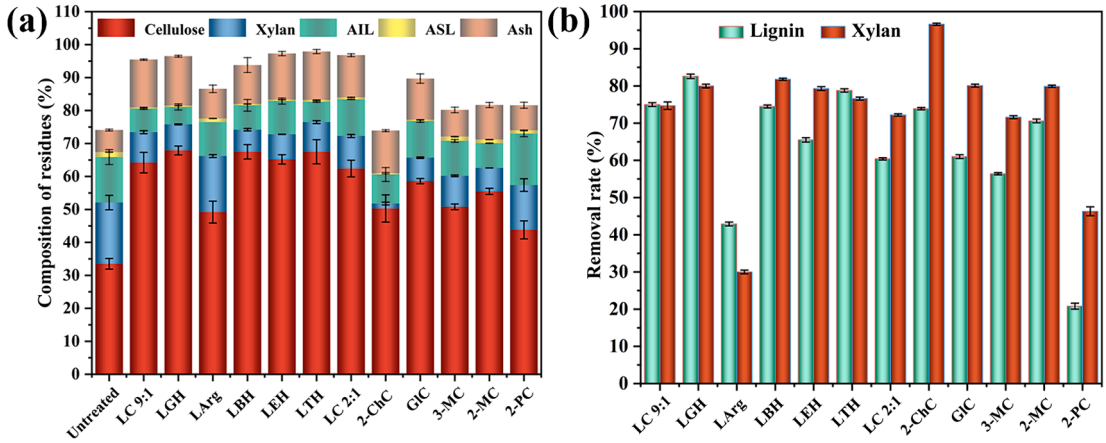


Fig. 1. (a) Composition of pulps after DES treatment, (b) Removal rate of lignin and xylan. All experiments were repeated at least in triplicates.

(81.8 % vs. 74.7 %). Recently, it was reported that the solubility of xylan was linearly correlated with Kamlet-Taft parameters β value which represents a solvent's ability to donate electron density to form a hydrogen bond with solute's protons (Zhang et al., 2023). This also suggest a stronger interaction between LBH and xylan of biomass than that of LC. Besides, the stronger acidity of LBH was also responsible for the higher xylan removal rate for the acid-sensitive property of hemicellulose (Huo et al., 2023). As for the DESs LEH and LTH, it appears that LEH prefer to remove xylan, while LTH was more benefit for delignification. A plausible explanation is that LEH with functional groups such as $–NH_2$ and –OH was likely to disrupt the intra-hydrogen bonding of hemicellulose by forming hydrogen bonds with –OH of hemicellulose, leading to slightly higher xylan removal rate as compared to that of LTH. In contrast, LTH containing more hydrophobic methyl group ($–CH_3$) was able to interact with hydrophobic lignin, and thus facilitated lignin dissolution. Wang et al. reported that the EmimOAc/ethanolamine-superbase mixture, as a promising cosolvent, could synergistically boost the dissolution of lignocellulose under relatively mild conditions through forming hydrogen bonds and an appropriate microstructure (Wang et al., 2023b). In addition, monoethanolamine was also proved to be a superior component of DESs when coupled with choline chloride to deconstruct poplar in other work (Yang et al., 2021). In brief, factors like alkyl chain length, functional groups, hydrophobicity and acidity of the HBA impact greatly on the pretreatment efficiency of DESs.

Focusing on HBDs which are known as a key factor in deconstruction of biomass, some monocarboxylic acids with different functional groups were also selected as components to coordinate with ChCl. As shown in Fig. 1b,when the molar ratio of lactic acid to ChCl was reduced to 2:1 from 9:1, the removal rate of xylan and lignin decreased slightly due to the reduced amounts of free protons and weakened hydrogen bonding interaction energy (Su et al., 2021). It is worth noting that replacing –OH of lactic acid by Cl^-, the removal rate of xylan and lignin enhanced significantly due to the strong electron-withdrawing effect of Cl^- (Wang et al., 2023a). Compared with LC, since the carbon chain length was shortened by one unit based on lactic acid, a slight increase of xylan removal (80.2 % vs. 72.3 %) was observed in GLC group. This phenomenon was coincident with the findings reported by Zhou and Wu that the xylan removal efficiency decreased with the increase of aliphatic chain length of monocarboxylic acid (Zhou & Wu, 2023). However, this is opposite to the rule that solubility of xylan positively correlated with the β values of the DESs since the higher β of LC was reported than that of GLC (0.55 > 0.50) (Farooq et al., 2020; Yu et al., 2022). It means that apart from Kamlet-Taft parameters, other factors should also be considered. Interestingly, the incorporation of sulfydryl group (-SH) instead of –OH caused the opposite effect depending on the site of -SH. It was found that the 3-position substitution (3-MC) caused a slightly decline in the lignocellulose fractionation efficiency, whereas the presence of 2-position -SH (2-MC) displayed an obvious better performance compared to LC. The reason is unclear, and it is worth exploring. In addition, when the –OH of lactic acid molecule was replaced by phenyl group, the pretreatment efficiency was greatly reduced, probably owing to the large extent of weakened hydrogen bonding capacity of 2-PC. Consequently, the lowest delignification of 20.7 % was observed.

3.2. Characterization of the rice straw pulps

To better understand the mechanisms and the effects of the HBD and HBA on the deconstruction of lignocellulosic matrix, the changes of the functional groups in the DESs fractionated pulps were analyzed by FTIR (see supplementary materials). The peak at 3400 cm^{-1} was assigned to the –OH stretching vibration of cellulose. This peak became broader for the pulps, suggesting the expose of cellulose and increase of the free –OH by means of removal of the amorphous components and disruption of the hydrogen bonds in cellulose. The peaks at 1512 cm^{-1} (aromatic skeletal vibration) and 1249 cm^{-1} (C–O vibration of syringyl) represented to lignin (Cheng et al., 2023), and the remarkable reduced signals were observed in most groups except for the pulps treated by LArg, 2-PC, 3-MC and GLC. The results confirmed the occurrence of lignin removal during the fractionation and agree well with the delignification trend of the DESs discussed above. Surprisingly, the intensity of the peak at 1730 cm^{-1} which corresponds to unconjugated C = O in hemicellulose did not significantly decrease, although the significant removal of hemicellulose and effective delignification occurred (as it is confirmed by Fig. 1b). On the contrary, the peaks even become stronger after pretreatment. This probably implies that some ester bonds in lignin or hemicellulose were kept during the processes. Nonetheless, these spectra peaks had the same trend as pretreatment efficiency (xylan and lignin removal ability) of these DESs and was consistent with the conclusion of the compositional analysis.

XRD patterns and the crystallinity index (CrI) (see supplementary materials) of the original rice straw and pulps showed diffraction peaks around 16°(1 0 1), 22°(0 0 2) and 34°(0 0 4), which are typical diffraction peaks of native cellulose I. The crystalline structure of cellulose did not change during the DESs pretreatment. The CrI value of the original biomass was 38.2 %, and the values of the cellulose-rich pulps increased to the range of 43.3 % to 57.6 %, indicating the removal of amorphous substances like hemicellulose and lignin (Cheng et al., 2023).

3.3. Enzymatic saccharification of pulps

There were many factors affecting to the enzymatic hydrolysis performances of cellulose in pulps, including the lignin and hemicellulose content and distribution, porosity, accessibility of cellulose, cellulose crystallinity etc. (Alokika et al., 2021). Fig. 2a shows the enzymatic digestibility for the polysaccharides of pulps. Overall, the saccharification efficiency improved significantly after DESs treatment. Generally, the higher the removal rate of lignin or xylan, the better the enzymatic degradation of carbohydrates. For instance, for LGH-involved case which has excellent delignification (82.5 %) and xylan removal (80.0 %), cellulose and xylan degradation reached 78.6 % and 93.0 %, respectively. Meanwhile, the pulp treated by LArg possessed the lowest glucose and xylose yields (34.6 % and 27.3 %, respectively), owing to the poor deconstruction performance. It was found that the 2-MC-fractionated pulp showed the maximum saccharification efficiency (93.3 % and 96.8 % for cellulose and xylan, respectively) regardless of the moderate degree of deconstruction (70.6 % and 79.9 % for lignin and xylan removal, respectively). The reasonable explanation was that the reductive nature of -SH hindered the over-degradation and condensation of lignin and xylan, which further minimized the formation of cellulase inhibitors. Notably, 2-ChC which caused considerable amounts removal of lignin and xylan (74.0 % and 96.5 %, respectively) only afforded moderate hydrolysis of cellulose (75.0 %) and xylan (44.7 %), being attributed to the excessive degradation of xylan and lignin and the formation of harmful substances for irreversible binding with cellulase (Hou et al., 2018).

Aiming to get insight into the relationship between the structure properties of pulps and performance of DESs, the removal rate of lignin and xylan were correlated with cellulose digestibility in saccharification stage (Fig. 2b). It was found that the rate of xylan removal (R^2 = 0.71) showed a stronger liner correlation than that of delignification rate (R^2 = 0.51), indicating that weakening of interaction between xylan and cellulose by these DESs are beneficial to the accessibility of cellulose to enzymes. As reported by Lin et al. (Lin et al., 2020), ChCl-Lac pretreatment of bamboo showed the same extent of positive correlation between xylan removal and enzymatic digestibility (R^2 = 0.71). Nevertheless, the correlation between lignin removal rate and cellulose digestibility established in this work is poorer than that of Lin's study (R^2 = 0.51 vs R^2 = 0.72). It was widely accepted that appropriate physicochemical features of substrate, particularly for the accessibility of cellulose, were mainly responsible for the satisfactory enzymatic hydrolysis of cellulose, and the considerable amount of xylan removal

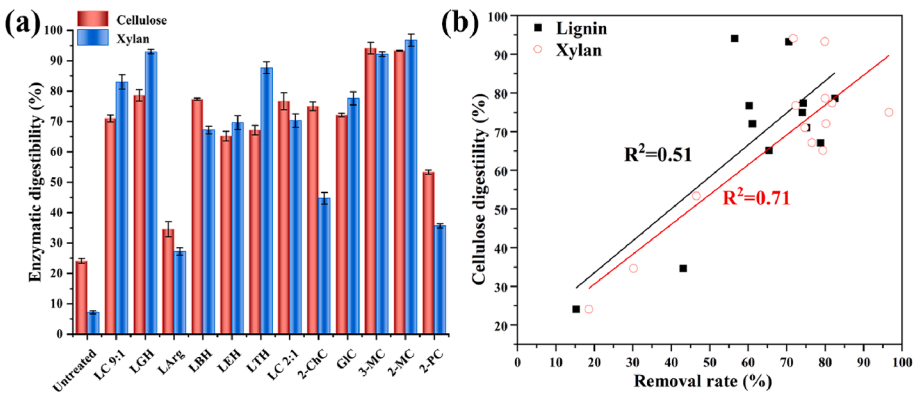


Fig. 2. (a) The enzymatic digestibility of the polysaccharides in the pulps, (b) the relationship between the removal rate of lignin and xylan and the enzymatic digestibility of cellulose.All experiments were repeated at least in triplicates.

could contribute to the enhanced cellulose accessibility to enzymes, thus achieving improved saccharification (Yang et al., 2021).

3.4. Characterization of lignin

The LRMs were isolated from the liquid stream after fractionation, apart from the 2-PC-involved case in which the dissolved lignin was hard to collect for characterization. The structures and properties of the recovered lignin were carefully analyzed. The compositions of lignin samples are shown in Fig. 3a. The recovery of lignin was less than 20 %, and the recovered lignin after DESs fractionation exhibited high purity (>80 %) except for the one extracted by LArg, suggesting that most of the lignin are valuable for further valorization (Zhou et al., 2022).

The Mw, Mn and the dispersity of DESs-fractionated lignin were measured by GPC (Fig. 3b). The *Mw* of the recovered lignin varied from 538 g mol^{-1} to 1249 g mol^{-1}, indicating the extensive depolymerization of lignin during the DESs fractionation, in particular for 2-ChC, 2-MC, 3-MC and LTH groups. Therefore, these DESs have strong ability to break down the inter-bonds of lignin, which was basically in line with the excellent delignification capability of them. Besides, Mw maintained in a range of 1110–1300 g mol^{-1} with low DPI (1.147–1.734) for most of the lignin specimens, illustrating the recovered lignin has mall and homogeneous structure (Alvarez-Vasco et al., 2016).

To compare the effect of HBD and HBA on the functional structures of lignin, the fractionated lignin specimens were further analyzed by FT-IR. The broad band appearing at 3360 cm^{-1} was attributed to the O-H vibration phenolic and aliphatic (–OH). The peak for C = O stretching (1730 cm^{-1}) existed in all the tested lignin specimens, indicating the

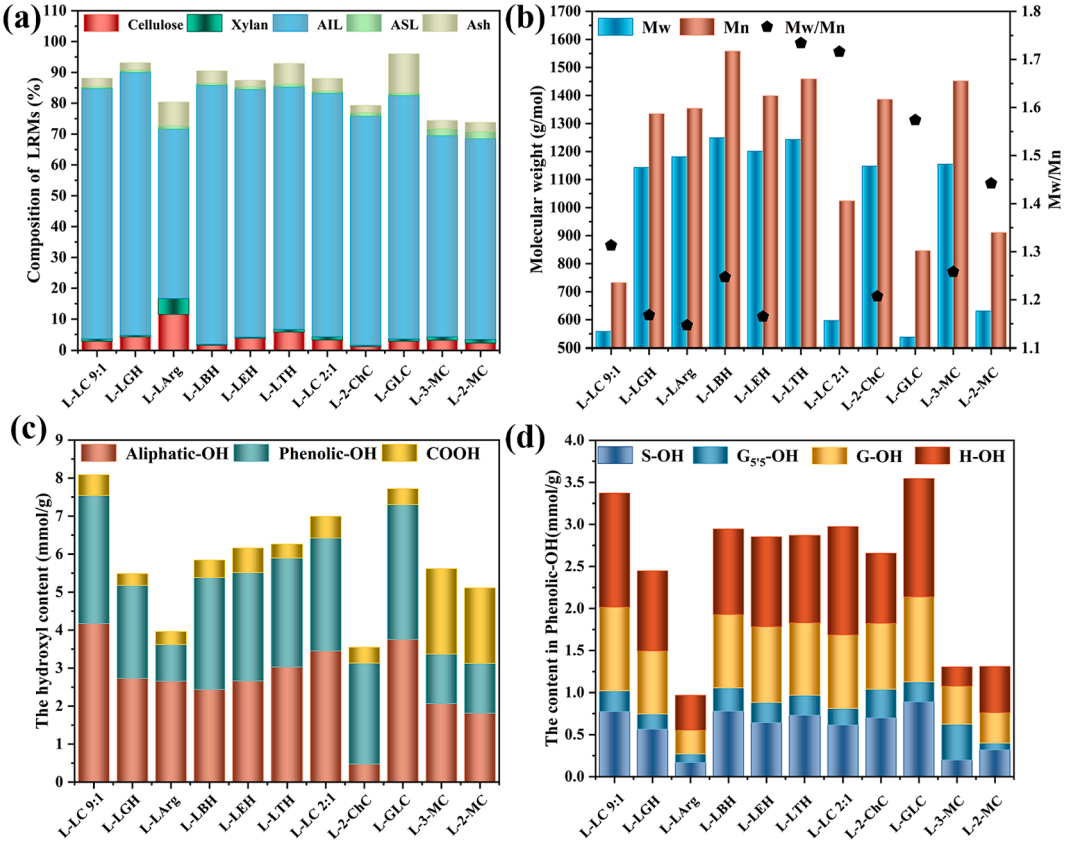


Fig. 3. (a) The composition of the LRMs, (b) The *Mw* and *Mn*, and the dispersity of the DES fractionated lignin, (c) The hydroxyl content of the LRMs, and (d) The content of different categories in Phenolic-OH.

presence of ester bond in lignin (Guo et al., 2022). The peaks at 1423–1655 cm^{-1} were attributed to typical aromatic skeletal vibrations, suggesting the well preservation of lignin skeleton (Guo et al., 2022). Notably, the obvious weak absorbance at 1514 cm^{-1} was observed in LArg-extracted lignin, which was attributed to the poorest depolymerization of lignin in LArg and the low purity of it. The signals at 1328 cm^{-1} and 1220 cm^{-1} were corresponding to stretching vibrations of Guaiacyl (G) type lignin. The band at 835 cm^{-1} was also attributed to the vibration of the G units (Melro et al., 2023). Meanwhile, all the lignin specimens exhibited similar spectra profiles, and there were no significant impurity peaks occurred.

The total –OH content was an important indicator for the structural changes of lignin. The –OH content of lignin specimens were determined by ^{31}P NMR (Fig. 3c). All the lignin specimens contained large amounts of total –OH, revealing the strong ether breaking ability of DESs. In particular, lignin isolated from the LC 9:1 possessed the highest total –OH content of 8.089 mmol g^{-1}, indicating the strongest ether bonds breaking ability of this solvent (Zhao et al., 2022). Besides, lignin isolated from the GLC was also observed with high total –OH content (7.722 mmol g^{-1}), whereas the lignin isolated from 2-ChC contained the lowest amount of overall –OH (3.548 mmol g^{-1}) due to the extensive cleavage of the ether bonds and partial condensation of the exposed –OH in this fierce solvent (Zhang et al., 2016). Unsurprisingly, low total –OH content (3.962 mmol g^{-1}) was observed for L-LArg due to the low extent of ether bonds cleavage caused by this solvent, which could further limit the formation of –OH (Su et al., 2021). The aliphatic –OH contents varied from 0.476 mmol $g^{-1}$to 4.173 mmol g^{-1}, representing 35.5–51.6 % of the total –OH for most of the lignin specimens, except for the 2-ChC and LArg involved processes. Similar to total –OH groups, it was found that lignin isolated by 2-ChC exhibited the lowest aliphatic-OH content (accounts for 13.4 % of the total –OH groups), while the phenolic-OH became dominant one (as high as 74.9 %). The increase of the phenolic –OH content can be attributed to the effective cleavage of β-O-4 bond or the demethylation of methoxy groups in lignin (Shen et al., 2020). Besides, the $G_{5'5}$-OH referencing the condensed lignin varied from 0.103 mmol g^{-1} (LArg) to 0.428 mmol g^{-1}(3-MC). It seems that the DESs with strong delignification capability always prefer to extract lignin with high $G_{5'5}$-OH content, indicating the occurrence of condensation during the fractionation (Hong et al., 2020). As is shown in Fig. 4d, L-3-MC showed the largest amount of $G_{5'5}$-OH (0.428 mmol g^{-1}), while L- 2-MC displayed the lowest $G_{5'5}$-OH content (0.081 mmol g^{-1}). Moreover, all the -SH group containing DESs showed low total –OH content, in which the phenolic OH (23–25 %) content was apparently lower than –COOH (~40 %) content. This may be due to the fact that the presence of -SH group in the DESs could promote the cleavage of the β-*O*-4 bonds, and form small molecular weight lignin fragments.

To further investigate the delignification mechanism of the candidate DESs, the 2D-HSQC NMR spectra were acquired. The assignment of the main cross-signals of the HSQC spectra of the aromatic (δ_C/δ_H 100.0–150.0 /9.0–5.5 ppm) and side-chain (δ_C/δ_H 50.0–90.0/2.50–6.0 ppm) regions of the lignin specimens (see supplementary materials) was referred to literatures (Chen et al., 2020). With respect to the side chain region ((a)-(k)), intensity signal of β-O-4 (A_α) at 68.36/4.93 ppm appeared in most of the lignin specimens, except for the ones fractionated by 2-ChC, 2-MC and 3-MC. Besides, β–β (B_α) and β-5 (Cα) bonds in all the lignin specimens exhibited extremely weak signals. The β-positions of β-O-4 (A_β), β–β (B_β), and β-5 (C_β) signals which normally center at δ_C/δ_H86.1/4.05 (A_β for S) and 83.4/4.33 (A_β for G), 53.5/3.05, and 52.4/3.45 ppm, respectively, were also almost disappeared. The γ-position of β-O-4 (Aγ), β–β (B_γ), and β-5 (C_γ) signals were identified at δ_C/δ_H 59.47/3.48 ppm, 69.01/3.64 ppm, and 64.21/3.53 ppm, respectively. Likewise, the γ-acetylated *β*-O-4 bonds (A'$_\gamma$) at 63.0/4.36 ppm were also clearly recognized in most of the lignin specimens (except for the 2-ChC, 2-MC and 3-MC fractionated lignin), indicating the existence of LCC structures (Chen et al., 2020). This is consistent with the composition analysis in which small amounts of hemicellulose and cellulose remaining in the fractionated lignins. The arabinose and xylose signals were observed at 65.43/4.19 ppm and 76.67/3.80 ppm, respectively.

The aromatic regions ((l)-(v)) provided the information about syringyl (S), oxidized syringyl(S'), G (G_2, G_5, G_6), and *p*-hydroxyphenyl (H) unit signals. The signals of *p*-coumaric acid (PCA) identified at δ_C/δ_H 113.58/6.33 ppm. The signals belonging to G units were at δ_C/δ_H 110.64/7.01 ppm (G_2), 115.53/6.79 ppm (G_5), and 118.55/6.79 ppm (G_6). The *p*-hydroxyphenyl (H) signal appeared at δ_C/δ_H 128.5/7.24 ppm ($H_{2,6}$). In addition, only a small amount of condensed S lignin was identified in the lignin specimens and the condensation of G unit was not obvious for most lignin. Therefore, most of the DESs used in this work could effectively protect the *β*-O-4 bonds in lignin structure. Moreover, condensation reactions mainly happened on G-units particularly for lignin isolated from 2-ChC and 3-MC, reflected by the more abundant G_{2cond} signals than those of others. Weak G_6 signal were observed for the 2-ChC, LEH and LTH fractionated lignin, implying the occurrence of some chemical modification of C_6-H_6 in G units (Chambon et al., 2019). Interestingly, PCA signals nearly disappeared for 2-MC, 3-MC and 2-ChC related cases, accompanying by the enrichment of the $H_{2,6}$ signal. This result suggests that the -SH or -Cl containing DESs convert *p*-coumaric acid into H type structures during the fractionation process (Chen et al., 2020). Besides, the -SH containing DESs could also effectively preserve the G type structure of lignin. As shown in the aromatic regions, more condensed G_2 signal was found for the 3-MC extracted lignin than that of 2-MC. This phenomenon agreed with the quantitative ^{13}P NMR data (the –OH groups analysis) (Fig. 3d).

The semi-quantitative analysis of typical structures of the

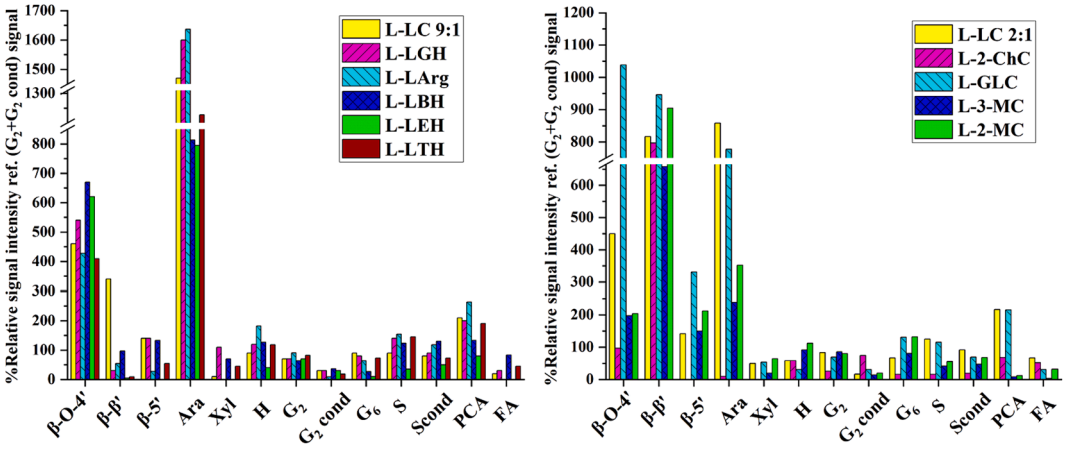


Fig. 4. The semi-quantitative analysis of typical structures of the fractionated lignin.

fractionated lignin was conducted, so as to clarify the structural differences by various types of DESs fractionation (Fig. 4). It was found that most of DESs kept abundant β-O-4 bond in the lignin specimens. As 2-ChC and 2-MC exhibited strong biomass deconstruction ability, and low amount of hemicellulose (arabinose and xylose units) were detected in the corresponding lignin specimens. Moreover, the remarkably higher content of H and lower PCA content were also observed in 3-MC and 2-MC extrated lignin. Additionally, extremely high content of G_{2cond} structure was observed for the 2-ChC extracted lignin. Overall, it is worth mentioning that most of the candidate DESs could well-preserve the β-O-4 bond, and protect the fractionated lignins from condensation. Therefore, the lignin specimens could be potentially valorized into valuable chemicals and materials in the downstream processes.

3.5. Bioethanol fermentation using enzymatic hydrolysate of the pulps

The above sections demonstrated that the fractionation efficiency of the lignocellulosic matrixes was dependent on the types and characteristics of HBD and HBA in DESs. The effective saccharification of the pulps drove to further investigate the possibilities of transformation of the DESs fractionated streams to valuable products. On this basis, this section took bioethanol fermentation from the DESs pulps as an example, and emphasized on the influence of HBD and HBA of DESs on the yield of products after biotransformation.

Batch bioethanol fermentation was carried out using the enzymatic hydrolysates of the rice straw pulps from the upstream DESs fractionation processes. *S. cerevisiae* YSX4C222, the xylose and glucose co-fermentation strain, was adopted. From the results of polysaccharide digestion, a maximized 87.3 ± 2.3 g/L of glucose and 12.3 ± 0.4 g/L of xylose (total 99.6 ± 2.7 g/L) were obtained from the LGH 9:1 treated pulps, followed by 2- MC (total sugar concentration of 96.5 ± 2.2 g/L) and 3-MC (total sugar concentration of 93.6 ± 1.3 g/L), while only 34.7 ± 2.3 g/L of monomer sugars can be produced from the LArg case under the similar conditions.

As for the batch fermentations, different to the negative influence of the degradation products of the pulp towards the enzymatic hydrolysis, there were no obvious inhibitions of the rice straw hydrolysates to the yeast cells. As shown in Fig. 5a and b, all of the monomer sugars in the initial substrates can be used up at the end of fermentation, no matter the DESs types in the fractionation stage, and the ethanol yield and productivity were maintained at 0.41–0.47 g g^{-1}monomer sugars and 0.17–0.55 g/Lh^{-1}, respectively. Due to the higher concentration of sugars in substrate, a higher bioethanol titer of 45.4 ± 0.6 g/L was acquired using the hydrolysate of LGH group (Fig. 5c). Besides, 3-MC and 2-MC fractionated pulps also afforded high ethanol concentration and productivity especially for 3-MC which has less condensation and inhibitors than 2-MC. Time course of the batch ethanol fermentation of the LGH group is shown in Fig. 5d. In accordance with the fermentation using the similar strain (Wu et al., 2023), the glucose consumption was faster than the metabolism of xylose although the monomer sugars could be used together by the YSX4C222 strain. Meanwhile, the productivity and yield of ethanol using the LGH 9:1 hydrolysate was only slightly lower than those of control group (the dotted lines in Fig. 5d) that using a synthetic medium with similar sugars concentration and constitution, indicating little inhibition of the by-products in the DESs hydrolysate.

Mass balance of the complete rice straw biorefinery process was analyzed, and the transformation routes relying on the LGH, LTH, LC (9:1), 2-ChC, 3-MC and 2-MC fractionations were compared (see supplementary materials). The structural resistance of the rice straw can be properly overcome by the effects of DESs, in which process the HBD and HBA were acted as important roles. The effective decomposition of the chemical structure of rice straw by DESs afforded high yield of fermentable sugars and highly activated technical lignins, and the fermentable sugars were further valorized into bioethanol. The mass balance of this biorefinery process indicated that the yields of bioethanol were sensitive to the types of DESs used in the pretreatment stage.

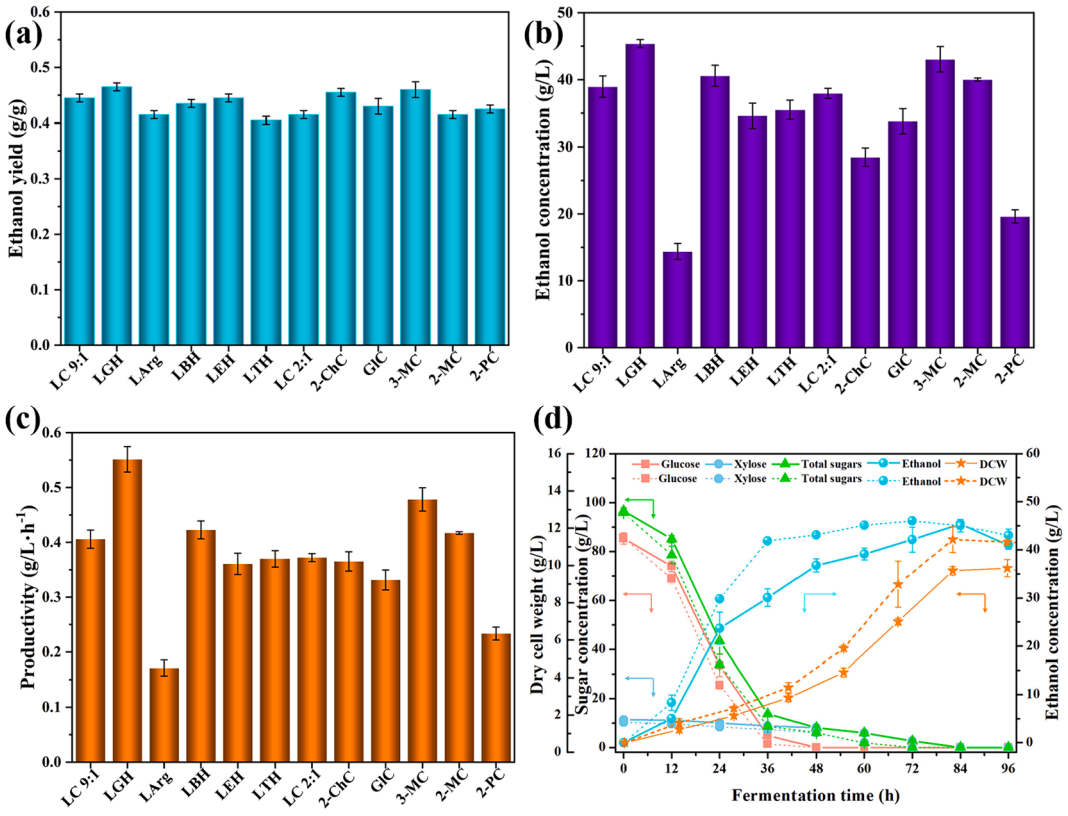


Fig. 5. (a) The ethanol yield, (b) concentration, (c) productivity after fermentation of the enzymatic hydrolysates of the DES pulps, and (d) the ethanol fermentation process for LGH 9:1 related case. All experiments were repeated at least in triplicates.

Compared with other candidate routes, the LGH fractionation group allowed 302.8 g of bioethanol production from 1 kg of dried rice straw, while the 119.0 g of the technical lignin by-products were harvested. 3-MC, 2-MC, LC 9:1 and LTH fractionation of rice straw allowed comparable amounts of valuable products, while 2-ChC had better performance in producing lignin.

4. Conclusions

A rice straw biorefining process based on DESs fractionation was studied. The results showed that the fractionation performances of rice straw were highly dependent on the structure and properties of the HBD and HBA of DESs, including acidity, alkyl chain length, hydrogen bonding ability, and functional groups, etc. Compared to other candidate DESs, LGH demonstrated superior performance in green biorefining for excellent delignification and xylan removal, 78.6 % cellulose hydrolysis, and 302.8 g of bioethanol/ kg rice straw.

CRediT authorship contribution statement

Pengfei Zhao: Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation. **Fulong Pu:** Validation, Methodology, Investigation, Conceptualization. **Changsheng Su:** Writing – review & editing, Visualization, Validation. **Yefan Wan:** Writing – review & editing. **Texin Huang:** Writing – review & editing. **Xuedan Hou:** Supervision, Resources, Funding acquisition. **Di Cai:** Supervision, Resources, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2024.130635.

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